INCH-POUND

MIL-W-80046D
9 March 1990
SUPERSEDING
MIL-W-80046C

MILITARY SPECIFICATION

# WELDING MACHINES, RESISTANCE, STORED ENERGY, CAPACITOR DISCHARGE, DUAL RANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers dual range, capacitor discharge, stored energy, resistance welding machines with equipment and accessories specified herein.

1.2 Classification. The welding machines shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 3.5 and 6.2).

Class A - Automatic and semi-automatic machines,
Class B - Manual and intermittent duty machines.

Size 11 - 50 watt-second capacity.
Size 12 - 50 watt-second capacity.
Size 13 . 100 watt-second capacity.
Size 14 - 125 watt-second capacity.
Size 15 - 250 watt-second capacity.
Size 16 - 500 watt-second capacity.
Size 17 - 750 watt-second capacity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DOOISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-451 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-5TD-1276 - Leads for Electronic Component Parts.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)



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29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001. )

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ANSI B93.114M - Pneumatic Fluid Power - Systems Standard for Industrial Equipment and Machine Tool Applications.

(Application for Copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3361.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council c/o The National Tool Builders Association 7901 Westpark Drive, McLean, Va. 22101-4269.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

RESISTANCE WELDER MANUFACTURERS ASSOCIATION (RWMA)

RWMA Bulletin Number 16 - Resistance Welding Equipment Standards. (DOD adopted)

(Application for copies should be addressed to the Resistance Welder Manufacturers Association, 1900 Arch Street, Philadelphia, PA 19103-1403.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)



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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete welding machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The welding machine shall be new (not a prototype] and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The class A machine shall be capable of semi-automatic and automatic operation. The class B machine shall be capable of performing manual and intermittent welding. All parts subject to wear, breakage or distortion shall be readily accessible for adjustment, replacement, or repair. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2], or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicatlng devices on the machine shall be graduated in the same system.

3.2.2 Reclaimed materials. The welding machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The welding machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator s work station (s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assembles furnished on the welding machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial welding machine suppl ied to the commercial market shall be provided for the point of operation and all other parts of the welding machine that present safety hazards (see 6.2).



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3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine, and its components materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The welding machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion, or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise Specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the welding machine shall be painted in accordance with the manufacturer’s commercial practice and color.



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3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than an enclosure, a power supply unit, a welding head, an electrical system, and electrodes and electrode holders.

3.4.1 Enclosure. The metal enclosure shall be constructed of corrosion resistant material or treated to resist corrosion, and shall have provision for inspection and maintenance. Ventilation and cooling shall be adequate to prevent temperatures in excess of design requirements under normal operating conditions at rated power output. Where ventilation is provided by filtered forced air, the air inlet shall be provided with replaceable filters. The sizes 6 and 7 welding machine power supply and controls may be enclosed 10 two separate units

3.4.2 Power supply unit. The power supply unit shall include the voltage regulator, capacitors, controls, lockout circuits and overvoltage protection devices. The primary line voltage shall pass through a rectifying circuit to the capacitors.

3.4.2.1 Voltage regulator. The voltage regulator shall maintain the capacitor charge voltage within $\pm$1 percent with a line voltage fluctuation of $\pm$ 10 percent.

3.4.2.2 Capacitors. The capacitors shall operate satisfactorily and maintain their measured rated microfarad value when operated at ambient temperature. Above ambient and up to 104 degrees Fahrenheit the capacitors shall maintain their measured rated microfarad value within $\pm$ 10 percent.

3.4.2.3 Charging lockout circuit. When specified (see 6.2), the power supply unit. shall have the means to prevent welding initiation until the capacitors are charged to within 1 percent of that selected on the weld energy range control.

3.4.2.4 Discharge lockout circuit. The power supply unit shall have the means to prevent capac itors from beingcharged until after they are discharged after a welding cycle.

3.4.2.5 Overvoltage protection. The power supply unit shall have an overvoltage protection device to prevent damage due to overvoltage.

3.4.2.6 Controls. The controls shall be adjustable for making repetitive welds as specified herein without readjustments. The controls shall consist of at least a dual range control, with a dual range indicator to expand the lower range scale readings, a weld energy control switch and dial, a "weld/no weld" control or master switch and a pilot light.



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3.4.2.6.1 Dual range switch. The dual range switch shall provide the means to change from a low welding range to a high welding range.

3.4.2.6.2 Weld energy range selector control. The weld energy range selector control shall operate to regulate the weld energy to the desired level in stepless adjustments. The control shall be marked in watt-seconds or percent of full rated watt-seconds. The power supply shall be equipped with a turn down circuit which allows the capacitor bank voltage to track the weld energy control when it is adjusted up or down.

3.4.2.6.3 Weld pulse selector switch. When specified (see 6.2), the welding machine shall be equipped with a weld pulse selector switch or internal jumpers for selecting two or more weld outlet pulse widths to the degree specified.

3.4.2.6.4 Automatic control adapters. When specified (see 6.2), the welding machine shall have provisions and be equipped with the proper adapters for connection to automatic controls such as timers, programmers, and NC equipment.

3.4.2.7 Main disconnect switch. The welding machine shall have a circuit breaker that will serve as an "on-off switch and as an overload protection device. Unless otherwise specified (see 6.2), a circuit breaker shall be furnished and shall be mounted on the front or rear control panel. A pilot light with a red lens shall be provided to indicate when the power supply is energized. The incoming electric power shall be fused when a circuit breaker is not furnished. When used as a protection device only, the fuse may be located on or inside the power supply unit.

3.4.2.8 Power supply cord. The power supply unit shall be furnished with not less than five feet of three-conductor flexible electric cord that conforms to NFPA 70, type SO. The cord shall terminate in an appropriate 3-pole grounding plug. The cord and plug shall be of the proper capacity, temperature, and voltage rating for the welding machine size specified. The cord shall have a strain relief connector where the cord enters the enclosure. The third wire shall be grounded to the frame of the unit.

3.4.2.9 Control circuit voltage. Unless otherwise specified (see 6.2), control circuit voltage shall be no greater than 24 volts.

3.4.3 Welding head. Unless otherwise specified, the welding head for class A welding machines shall be of the pneumatically operated type and the welding head for the class 5 welding machines shall be of the manually operated type. When specified (see 6.2), the welding head for class A welding machines shall be of the electrically or hydraulically, or pneumatically operated type and the class B welding machine shall be of the electrically, hydraulically, or pneumatically operated type.

3.4.3.1 Electrode force switch. The welding head shall have a switch to prevent the initiation of the welding cycle until the predetermined electrode force has been established.



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3.4.4 Electrical system. Unless otherwise specified (see 6.2), the electrical system shall conform to NFPA 70. Each welding machine shall operate on single phase, 60 Hz source, at either 115 or 230 volts. Unless otherwise specified (see 6.2), the sizes 11 through 16 welding machines shall be initially wired for operation on 115 volts and size 17 welding machine shall be initially wired for operation on 230 volts.

3.4.4.1 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.5 Electrodes and electrode holders. Unless otherwise specified (see 6.2), the electrodes and electrode holders shall be made of an alloy conforming to the requirements of RWMA Bulletin Number 16. Unless otherwise specified, the electrode holders for the sizes 1 through 6 welding machines shall be furnished for use with 1/8 inch diameter electrodes. When specified (see 6.2), the sizes 4 through 6 welding machines shall be furnished with electrode holders for use with 1/4 inch diameter electrodes. The size 7 electrode holders shall be for use with 1/4 inch diameter electrodes. The quanity, type, and size of electrodes required shall be as specified (see 6.2).

3.4.5 Pneumatic system. The pneumatic system shall conform to the requirements of ANSI 893.114M.

3.4.7 Hydraulic system. The hydraulic system shall conform to or exceed tne requirements of JIC H-1 and shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, provisions for draining, and a cleanout access plate.

3.4.8 Hour meter. When specified (see 6.2), the welding machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating tlme of the transformer. The meter shall be of the nonresetting type and shall have a range of O to 9,999 hours in increments of not greater than 1 hour, The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the welding machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the welding machine shall meet the requirements of table I for the welding machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or qreater.

3.5.1 Physical size limitations. Physical size of the machine shall be limited, when specified (see 6.2). The physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in Its future operating location.



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TABLE I. Machine size and capacities.

| Machine size | Watt-second weld energy capacity | Millisecond discharge time allowance | Intermittent repetition rate/minute 1 minute on 1 minute off | Material weld capacity 1/ (combined thickness in inches) | |
|---|---|---|---|---|---|
| | | | | Crosswire | Sheet/foil ribbon |
| 11 | Low range<br>.25 to 10<br>High range<br>.5 to 50 | <br>0.25 to 2<br><br>0.25 to 3.5 | <br>180<br><br>60 | <br>.017<br><br>.035 | <br>.004<br><br>.015 |
| 12 | Low range<br>.25 to 10<br>High range<br>.5 to 60 | <br>0.25 to 2<br><br>0.25 to 3.2 | <br>170<br><br>120 | <br>.019<br><br>.045 | <br>.005<br><br>.020 |
| 13 | Low range<br>0.4 to 20<br>High range<br>0.7 to 100 | <br>0.4 to 4<br><br>0.75 to 6.5 | <br>180<br><br>60 | <br>.019<br><br>.060 | <br>.006<br><br>.023 |
| 14 | Low range<br>0.4 to 20<br>High range<br>0.7 to 125 | <br>0.4 to 2<br><br>1 to 4 | <br>125<br><br>62 | <br>.019<br><br>.060 | <br>.006<br><br>.023 |
| 15 | Low range<br>2 to 30<br>High range<br>0 to 250 | <br>0.25 to 4.5<br><br>1.5 to 11.6 | <br>90<br><br>60 | <br>.035<br><br>.070 | <br>.012<br><br>.030 |
| 16 | Low range<br>2 to 50<br>High range<br>10 to 500 | <br>0.50 to 5<br><br>1.5 to 9.2 | <br>75<br><br>75 | <br>.035<br><br>.075 | <br>.015<br><br>.040 |
| 17 | Low range<br>4 to 100<br>High range<br>80 to 750 | <br>0.75 to 11<br><br>1.5 to 17 | <br>66<br><br>20 | <br>.060<br><br>.080 | <br>.023<br><br>.050 |

1/ MIL-STD-1276, type N for chemical composition only.

NOTE: Requirements are not less than shown.
Where a range is shown, the requirements are from the smaller figure or less to the larger or more.



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3.6 Performance. The welding machine shall be capable of producing the watt-second weld energy range specified in table I and produce electrical resistance welds on materials with dimensions and at the repetition rate specified in table I.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial welding machine shall be furnished (see 6.2).

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on the machine including all Instruments, control panels, control units, charts, plates, and related attachments shall be in the English language, Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each welding machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as Iisted below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each welding machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture



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3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2 ), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the welding machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements , however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b . Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.1.1.

4,4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The welding machine shall pass the examination and all tests to be accepted.



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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each welding machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each welding machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. Each welding machine shall be operated at no load for not less than 15 minutes. Proper operation of all controls, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The machine shall be adjusted to its maximum rated weld energy capacity and its rated repetition rate per minute and operated continuously for 2 hours of simulated welds. There shall be no overheating, arcing, or malfunction of the machine. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

4.7.3 Welding test. Spot welds shall be made using the minimum and the maximum size material in crosswire and sheet/foil specified in table I. Thirty consecutive welds shall be made in both size mater ials at the high and low weld energy capacity range and at the repetition rate specified in table I

4.7.3.1 Weld inspection. Five weld specimens each of the high and low weld energy capacity range and at the minimum and maximum size material required for the sheet/foil test, ana the crosswire test. shall be examined by 30 power magnification. There shall be no evidence of cracks or lack of fusion.

4.7.3.2 Weld strength test. After the welds have been inspected, they shall be subjected to a strength test by pulling or peeling the two pieces apart. There shall be no separation of the weld nugget.

4.7.4 Additional performance tests. Performance terts required in addition to or in lieu of the requiremdnts of 4.7.2 shall be as specified and fully described (see 6.2).

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The equipment covered in this specification is intended for use in contractor's plants, and military and government repair facilities. It is used for making connections in the circuits of semiconductors, integrated circuits, welding of wire and ribbons for transistor whiskers, strain gages, magnet wire, detonators, miniature electronic tubes, various thermocouple wire combinations, and hermetic sealing of semiconductors, rectifiers, crystal cases, and transformers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and Indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Painting, if different. (see 3.3.5.).

i. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

j. Charging lockout circuit, if required (see 3.4.2.3).

k. Weld pulse selector switch, if required (see 3.4.2.6.3).

l. Automatic control adaptors, if required (see 3.4.2.6.4).

m. Main disconnect switch, if different (see 3.4.2.8).

n. Control voltage, if different (see 3.4.2.10).

o. Welding head, if different (see 3.4.3).



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p. Electrical system, if different (see 3.4.4).

q. Electrical input power supply, if different (see 3.4.4).

r. Electrodes and electrode holders alloy, if different (see 3.4.5)

s. Size 4 through 6 electrode holder size, if different (see 3.4.5)

t. Electrodes, quanity, type, and size required (see 3.4.5).

u. Hour meter, if different (see 3.4.8).

v. Size and capacity, if different (see 3.5).

w. Additional performance features, if required (see 3.6.1).

x. Standard equipment, if different (see 3.7).

y. Optional equipment, if required (see 3.8).

z. Specify applicable repair parts, if required (see 3.9).

aa. Lubrication chart or plate, if required (see 3.10. 1).

bb. Nameplate, if different (see 3.10.2).

cc. Hydraulic requirements (see 3.10.3).

dd. Hydraulic plate (see 3.10.3).

ee. Inspection conditions, if different (see 4.3).

ff. First article inspection, if different (see 4.4).

gg. Quality conformance inspection, if different (4.5).

hh. Additional performance tests, if required (see 4.7.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

| Reference Paragraph | DID Number | DID Title | Suggested Tailoring |
|---|---|---|---|
| . | . | . | . |
| . | . | . | . |
| . | . | . | . |



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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the remarks section of the DD Form 1423 that all technical data furnished should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

| MIL-W-80046C | MIL-W-80046D |
|---|---|
| 1 - Deleted | |
| 2 - Deleted | |
| 3 | 11 |
| | 12 - New |
| 4 | 13 |
| 5 - Deleted | 14 - New |



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| | |
|---|---|
| | 15 - New |
| 6 | 16 |
| 7 - Deleted | 17 - New |

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

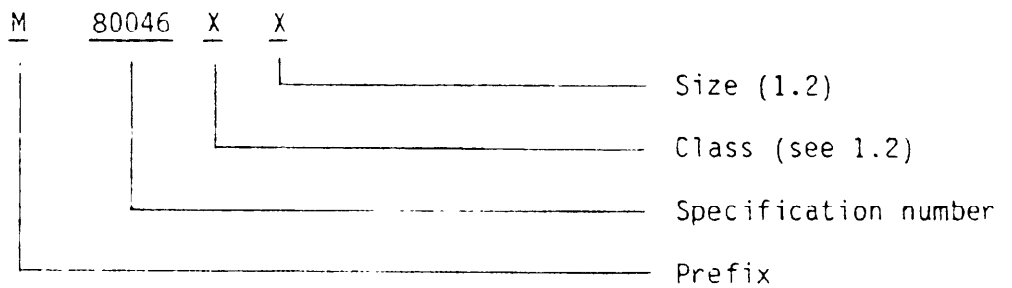
6.9 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PIN to be used for welding machines acquired to this specification are created as follows:




NOTE: (See MIL-STD-961C para. 4.4a)

6.14 Subject term (keyword) listing.

Electrode
Energy
Power supply
Transformer
Voltage regulator



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Custodians:
Army - AL
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - AR
Navy - MC, AS

Preparing activity:
DLA - IP
Project (3432-0143)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-W-80046D | 2. DOCUMENT DATE (YYMMDD)<br>90 03 09 |
|---|---|---|

3. DOCUMENT TITLE
Welding Machines, Resistance, Stored Energy, Capacitor Discharge, Dual Range

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

| 6. SUBMITTER | | |
|---|---|---|
| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

| 8. PREPARING ACTIVITY | |
|---|---|
| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial (2) AUTOVON |
| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

DD Form 1426, OCT 89

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